

Google Play Store App Insights

Power BI Data Analytics Project Report

Executive Overview & App Performance and Engagement

1. Executive Summary

This project analyzes Google Play Store application data to understand marketplace structure, app popularity, user engagement, ratings, pricing, app size, content ratings and update patterns. The final Power BI solution is intentionally presented through two dashboards: an Executive Overview and an App Performance & Engagement dashboard.

The source app dataset contains 10,841 rows and 9,660 unique app names after repeated app records are consolidated. Rating analysis uses 8,196 app records with valid 0–5 ratings. The review source contains 64,295 review records.

2. Business Objective

The case study asks the analyst to identify factors associated with app success, understand user preferences and behavior, and translate the findings into practical recommendations for app development, marketing and product management. The supplied brief identifies developers, product managers, marketing teams and senior management as key stakeholders.

3. Data & Preparation

Item	Result
Source app records	10,841
Unique app names	9,660
Exact duplicate rows	483
Repeated app-name rows	1,181
Valid rating records	8,196
Unique categories	34
Review records	64,295

- Missing values were reviewed; rating calculations exclude invalid and missing rating values.
- Installs, Reviews and Price were converted to numeric forms for analysis.
- Last Updated was converted to a date field.
- App size was converted to MB when a numeric size was available.
- Repeated app names were consolidated for app-level analysis.

4. Key Findings

Overall rating

The average valid app rating is approximately 4.17/5, indicating generally strong reported satisfaction.

Category diversity

The dataset covers 34 categories, providing broad coverage of the app marketplace.

Free vs paid

Free apps overwhelmingly dominate the dataset and average far more reviews than paid apps.

Demand concentration

Game, Communication, Tools, Productivity and Social are the top five categories by summed install tiers.

Engagement concentration

Facebook, WhatsApp Messenger, Instagram, Messenger and other major apps dominate review volume.

Rating vs popularity

Install count and rating have a weak positive correlation of about 0.04; popularity is therefore not a strong standalone predictor of satisfaction.

Pricing

Among paid apps, price and rating have a weak negative correlation of about -0.11.

App size

Size and installs have a weak positive correlation of about 0.13.

Updates

2018 accounts for about 66.5% of dated valid app records, showing a strong concentration in the dataset.

Sentiment

Among labeled reviews, approximately 64.1% are positive, 22.1% negative and 13.8% neutral.

5. Category & Engagement Analysis

Top categories by summed install tiers:

Category	Install tiers
GAME	13.88B
COMMUNICATION	11.04B
TOOLS	8.00B
PRODUCTIVITY	5.79B
SOCIAL	5.49B

Top reviewed apps:

App	Reviews	Rating	Installs
Facebook	78.2M	4.1	1B
WhatsApp Messenger	69.1M	4.4	1B

Instagram	66.6M	4.5	1B
Messenger – Text and Video Chat for Free	56.6M	4.0	1B
Clash of Clans	44.9M	4.6	100M

6. Answers to the Case-Study Questions

Basic-Level Questions

1. Average rating of apps

The average valid app rating is **4.17 / 5**. Overall satisfaction is therefore relatively strong, although this should be interpreted alongside the large number of missing ratings.

2. Unique app categories

There are **34 unique categories**. The dataset covers a broad range of app types, with Family, Game, and Tools among the largest categories by app count.

3. Distribution of app sizes

Among apps with a usable numeric size, **25.8%** are **≤5 MB**, **18.7%** are **5–10 MB**, **17.6%** are **10–20 MB**, **26.0%** are **20–50 MB**, and **11.8%** are **50–100 MB**. The median size is **12 MB**. In addition, **12.7%** of unique app records report 'Varies with device' rather than a numeric size.

4. Free vs paid apps

The market is overwhelmingly free: **7,592 valid free apps** versus **604 valid paid apps**. Free apps dominate the dataset, indicating that paid apps represent a relatively small segment.

5. Most common content rating

Everyone is the dominant content rating with **6,618 valid-rated apps**, far ahead of the other content-rating groups.

6. Top 5 most installed apps

The highest install tier is **1 billion+ installs**. Because many apps are tied at that tier, a strict top-five ranking is not unique. Using review count as a tie-break, the five leading apps are **Facebook, WhatsApp Messenger, Instagram, Messenger – Text and Video Chat for Free, and Subway Surfers**.

7. Apps rated 4.0 and above

6,286 apps have ratings of at least 4.0, representing **76.7%** of apps with valid ratings. High ratings are common in the dataset.

8. Average reviews: free vs paid

Free apps average about **274,690 reviews**, compared with about **10,919 reviews** for paid apps. The much higher review volume for free apps reflects their much larger market presence and user base.

9. Average app size by category

Game apps have the largest average size at about **42.9 MB**, followed by Family (**29.0 MB**) and Sports (**26.4 MB**). Tools has the smallest average among categories at about **8.3 MB**. This suggests category functionality is associated with substantially different storage requirements.

10. Apps last updated in 2018

5,450 apps in the valid app-level dataset were last updated in 2018, representing about **66.5%** of dated records. The dataset is heavily concentrated in 2018.

Medium-Level Questions

1. Correlation between installs and rating

The Pearson correlation is approximately **0.04**, indicating a **very weak positive relationship**. Install volume alone does not strongly explain rating. This supports treating popularity and satisfaction as related but distinct dimensions.

2. Categories with highest average rating

The strongest category averages are **Events (4.44)**, **Education (4.36)**, **Art & Design (4.36)**, **Books & Reference (4.34)**, and **Personalization (4.33)**. Category differences are relatively modest, so sample size should be considered before treating small gaps as decisive.

3. Effect of price on average rating

Among paid apps, price and rating show a **weak negative correlation of -0.11**. In broad price bands, apps priced at $\leq \$1$ average about **4.30**, while apps above \$30 average about **3.91**. Higher pricing therefore does not appear to guarantee better user satisfaction.

4. Rating distribution across content ratings

Average ratings are fairly close across content-rating groups. Among groups with meaningful sample sizes, **Everyone 10+ (4.23)** and **Teen (4.23)** are slightly above **Everyone (4.17)**, while **Mature 17+ (4.12)** is lower. The very small Adults only group should not be overinterpreted.

5. Genres with most apps above 1M installs

The leading genres are **Tools (172 apps)**, **Action (128)**, **Photography (123)**, **Communication (99)**, and **Productivity (91)**. These genres have the strongest representation among high-install apps.

6. App update frequency / average time between updates

The supplied dataset contains only one **Last Updated** value per app, not a sequence of historical update dates. Therefore, a defensible average time between consecutive updates **cannot be calculated from this dataset alone**. The data can support update-year analysis, but not true inter-update frequency.

7. Impact of app size on installs

The correlation between numeric app size and installs is about **0.13**, a weak positive relationship. Size therefore has little standalone explanatory power for download volume; distribution, category, utility, brand, and marketing are likely more important.

8. Apps with highest review counts and their ratings

The most reviewed apps are led by **Facebook (78.2M reviews, 4.1 rating)**, **WhatsApp Messenger (69.1M, 4.4)**, **Instagram (66.6M, 4.5)**, **Messenger (56.6M, 4.0)**, and **Clash of Clans (44.9M, 4.6)**. Very high review counts are concentrated among globally popular apps.

9. Content rating distribution: free vs paid

Both free and paid apps are dominated by **Everyone** content rating. Within each type, the share is approximately **81.4%** for free apps and **86.6%** for paid apps. Paid apps therefore have a slightly stronger concentration in the broad Everyone audience.

10. Top 5 categories by installs

The top categories by summed install tiers are **Game (13.88B)**, **Communication (11.04B)**, **Tools (8.00B)**, **Productivity (5.79B)**, and **Social (5.49B)**. These categories combine large user demand with strong download reach.

Advanced-Level Questions

1. Top 10 highest-rated apps and comparison of reviews/installs

The dataset contains many apps at the maximum rating of **5.0**, but many of these have very small review and install counts. This is an important analytical finding: **a perfect rating is not equivalent to market success**. Highly reviewed leaders such as Facebook, WhatsApp and Instagram have slightly lower ratings but vastly greater engagement and reach.

2. App update trend over time

Updates are heavily concentrated in **2018**, with **5,450** valid app records last updated that year. Earlier years decline steadily (for example, 2017 has 1,426 and 2016 has 628). This suggests the dataset is strongly weighted toward recent updates rather than showing a complete historical update cycle.

3. Average rating by install band

Average rating generally rises from lower-install bands toward the middle/high-install bands: **4.11** ($\leq 10K$), **4.10** (10K–100K), **4.20** (100K–1M), **4.27** (1M–10M), **4.36** (10M–100M) and **4.30** (100M–1B). This suggests moderately popular apps often have stronger ratings, but the relationship is not perfectly linear.

4. Sentiment analysis of app reviews

The review dataset contains **64,295** review records, of which **37,432** have a labeled sentiment: **Positive 64.1%**, **Negative 22.1%**, and **Neutral 13.8%**. Average polarity is **+0.37** for Positive, **-0.26** for Negative, and **0.00** for Neutral. Text signals frequently mention usability/ease, updates, ads, fixes, and gameplay. A full theme model would require explicit NLP topic extraction beyond the dashboard-level sentiment classification.

5. Relationship between genre and user ratings

Among genres with at least 10 rated apps, **Casual;Brain Games (4.48)**, **Events (4.44)**, **Puzzle;Brain Games (4.37)**, **Puzzle (4.37)** and **Education;Pretend Play (4.36)** are among the highest-rated. At the lower end are **Educational (3.87)**, **Dating (3.97)**, **Maps & Navigation (4.04)**, **Tools (4.04)** and **Trivia (4.04)**. Genre-level differences exist, but they should be interpreted with sample sizes in mind.

7. Business Recommendations

Prioritize high-demand categories

Use install concentration in Game, Communication, Tools, Productivity and Social to guide opportunity sizing and competitive research.

Balance quality with scale

Track installs, reviews and ratings together. A five-star app with very few reviews may not be commercially comparable with a 4.4-star app with tens of millions of reviews.

Optimize product footprint

Large categories such as Games and Family have higher average app sizes, so performance and storage efficiency should be part of product planning.

Use value-based pricing

Because price does not have a strong positive association with rating, paid products should justify price through functionality, quality and user value.

Operationalize review feedback

Monitor sentiment and recurring complaint themes, especially around ads, updates, bugs, usability and gameplay.

Respect data limitations

The Last Updated field records the latest update only. A historical update cadence requires multiple update events per app.

8. Dashboard Design

The final Power BI solution uses two dashboards. The Executive Overview is designed for rapid management-level scanning, while App Performance & Engagement provides category and app-level detail. The separation avoids unnecessary duplication and keeps the analytical flow concise.

Dashboard 1 — Executive Overview

High-level KPIs and overall marketplace performance.

Dashboard 2 — App Performance & Engagement

Category-level engagement, rating and installation analysis with interactive filtering.

9. Limitations

- The Google Play Store dataset is a historical snapshot rather than a live marketplace feed.
- Installs are provided in rounded tiers (for example, 1,000+ or 1,000,000+), so summed installs are estimates based on the supplied tier values rather than exact download counts.
- The Last Updated field contains one latest-update date per app and cannot provide a true average time between successive updates.
- Rating analysis is affected by missing ratings, and very high ratings can occur on apps with small review counts.
- The review table contains sentiment labels, but a rigorous theme analysis would require dedicated NLP/topic modelling.

10. Conclusion

The analysis demonstrates that Google Play Store app success is multidimensional. Strong market reach, engagement and user satisfaction do not always appear in the same apps or categories. The most actionable approach is therefore to combine installs, reviews, ratings, pricing, category demand, app size and review sentiment into a single decision framework.

Prepared as a Power BI portfolio project.